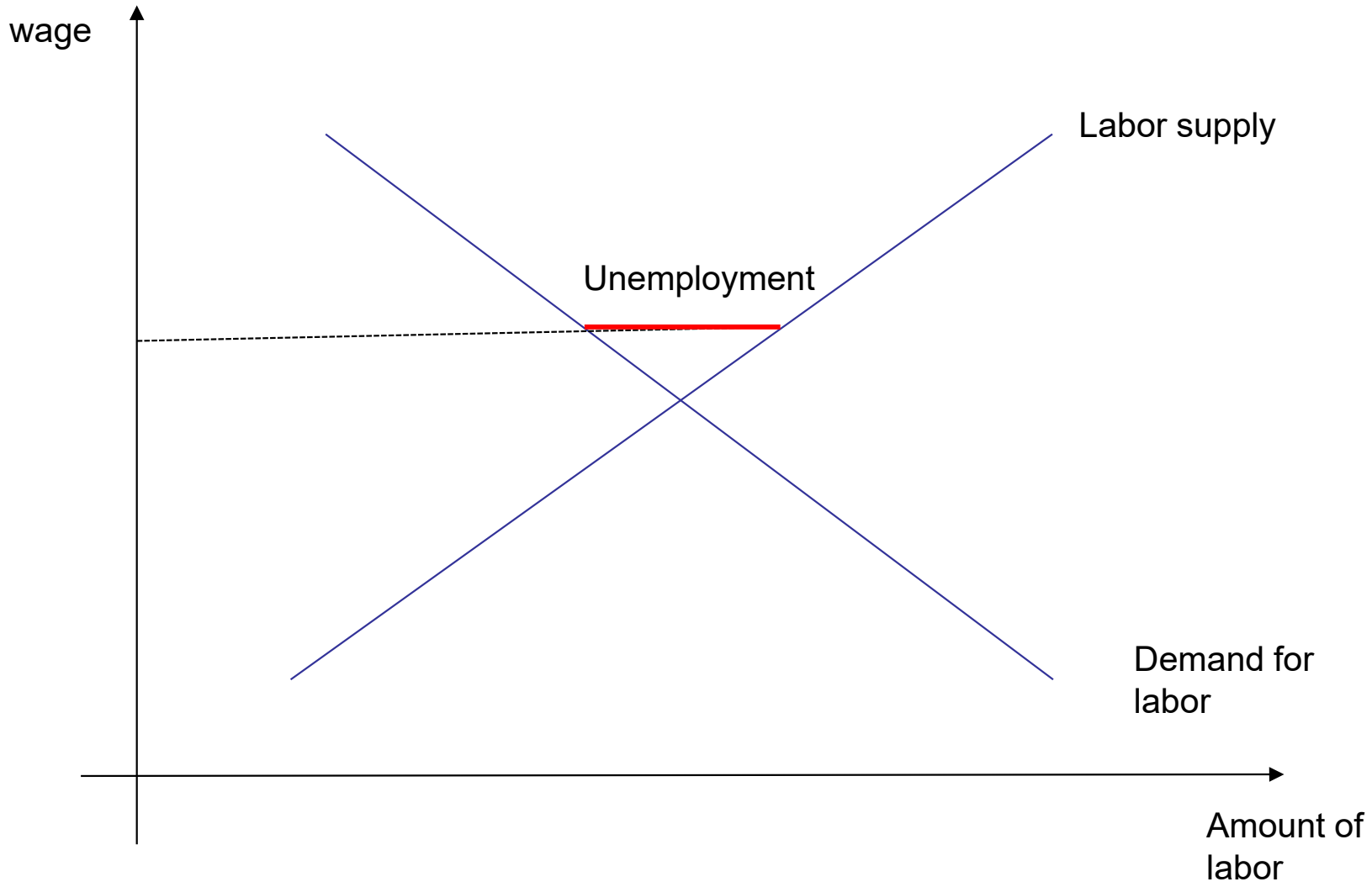
Putting social motives to work: Applications

Michal Bauer





Teaser: Why do we observe downward nominal rigidity in wages and thus unemployment?



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What are the explanations of downward nominal rigidity in wages and thus existence of unemployment? [writing some keywords is enough]

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Today

- **Social motives in organizations: Theoretical framework**
- Social motives in the labor market
  - Can employers motivate employees to exert more effort by giving them higher wages? Do wage cuts reduce effort? Does nature of the gift matters?
  - Inequality in the workplace: Comparison with peers
- Reciprocity in charitable giving: Can NGOs mobilize donations by distributing gifts?

How social motives influence workers' effort? Theoretical framework

- Standard **principal-agent model**
  - Two actors: a principal (organization) and an agent (the employee).
  - Principal hires the agent to produce output.
  - Output increases in the effort according to $f(e),\ where\ f' > 0\ and\ f'' < 0$
  - Effort is costly: $d(e), d' > 0, d'' > 0$

- **Moral hazard problem**
  - Arises because effort is unobservable and output depends on exogenous shocks (in addition to effort), thus employer cannot map output levels to effort
    - $output = f(e) + \varepsilon$, with $\mathrm{E}[\varepsilon] = 0$
  - Misalignment of interest between organization and employees: effort desirable for organization but costly for employees

How social motives influence worker's effort? Theoretical framework

Contract without social incentives

- Contract (total payment to worker): $y(e) = w + bf(e)$,
  - where w is a fixed wage and $bf(e)$ is a bonus tied to performance
- Principal chooses (w, b) to maximize:

  $(1 - b)f\big(e^*(w, b)\big) - w$, where e^* solves agent's utility maximization problem
- Agents/workers problem:

$$max_e U = w + bf(e) - d(e)$$

FOC: $b\frac{\partial f(e)}{\partial e} = \frac{\partial d(e)}{\partial e}$ (marginal financial returns to effort = marginal costs of effort)

Implication

Agent is solely motivated by performance pay and thus: whenever b=0, then $e^* = 0$

How social motives influence worker's effort? Theoretical framework

Introducing social motives $S(e_i)$

- Modified utility function of the agent: $U_i = w + bf(e_i) - d(e_i)$**$+S(e_i)$**
- FOC: $b\frac{\partial f(e_i)}{\partial e_i} + \frac{\partial S(.)}{\partial e_i} = \frac{\partial d(e_i)}{\partial e_i}$
- The **second term** captures **"social motives"** (marginal social reward of effort): it can be positive or negative, and thus social motives can stimulate and reduce optimal levels of effort

Implications: Some examples

- Altruistic preferences, such that $S_i = \sigma_{ij} U_j(e_i)$
  - Increase effort if effort creates benefits to a given group (e.g., employees identify with a mission of their organization)
  - Reduce effort if it creates harm to a group they care about (e.g., relative pay, negative externality on other employees)
- Reciprocal preferences
  - Effort increases when employer is kind, effort decreases when employer unkind
- Inequality aversion, such that $S_i = \sigma_{ij}|y_i - y_j|, \sigma_{ij} < 0$
  - Relative position matters: reducing effort if outcome is above norm, but increasing effort it is below

Today

- **Social motives in the labor market**
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Reciprocity and wage-effort hypothesis

Behavioral explanation:

- If workers respond to wages that are above a reference point by higher effort levels (positive reciprocity) and to low wages by lower effort levels (negative reciprocity), then this effects can explain why firms may want to pay more than market-clearing wage

- "On the worker's side, the gift given is work in excess of the minimum work standard; and on the firm's side the gift given is wages in excess of what these women could receive if they left their current job" (Ackerlof 1982)

- Laboratory evidence suggest that reciprocal preferences are very strong, and thus providing wages above market equilibrium might be profit maximizing.

Positive reciprocity on the labor market

Gneezy & List (2006)

- Motivation
  - Can reciprocal motives be used to stimulate effort on the labor market?
  - Duration of the task: hot responses vs. cold responses

- Task 1
  - Computerizing books in a library
  - Poster offer: $12/hour, 6 hours
  - Treatments
    - Control: $12/hour as promised
    - Treatment: $20/hour (surprise)

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What effects do you think the authors found? How do people respond to a surprise increase in their wages?

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Positive reciprocity on the labor market

Gneezy & List (2006)

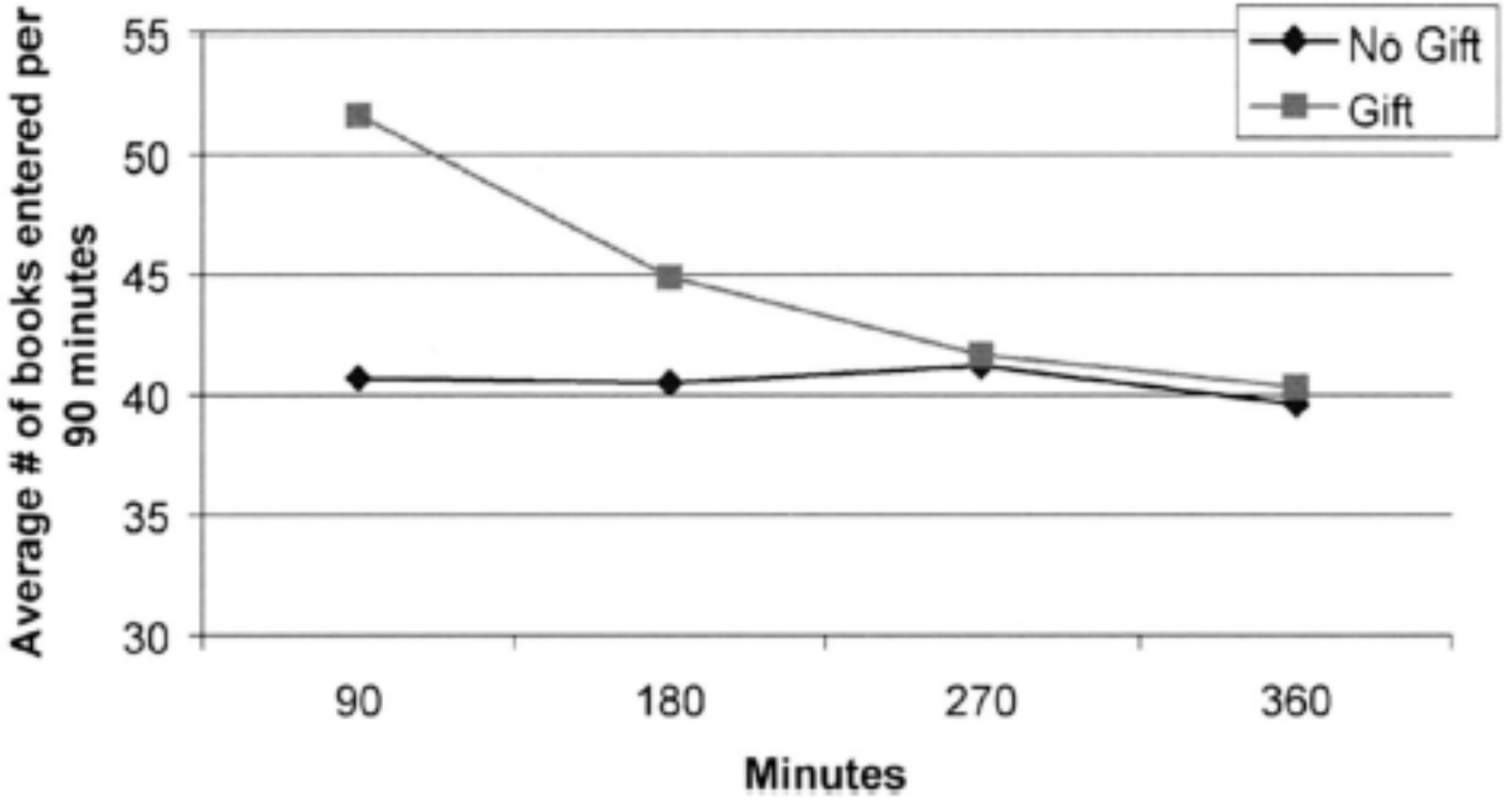


FIGURE 1.—Average books logged per time period.

Positive reciprocity on the labor market

Gneezy & List (2006)

- Task 2
  - Door-to-door fundraising campaign of an NGO
  - Poster offer: $10/hour, 2 days, prior training
  - Data: amount of contributions over time
  - Treatments
    - No-gift: $10/hour
    - Gift: $20/hour (surprise at the beginning of fundraising)

Positive reciprocity on the labor market

Gneezy & List (2006)

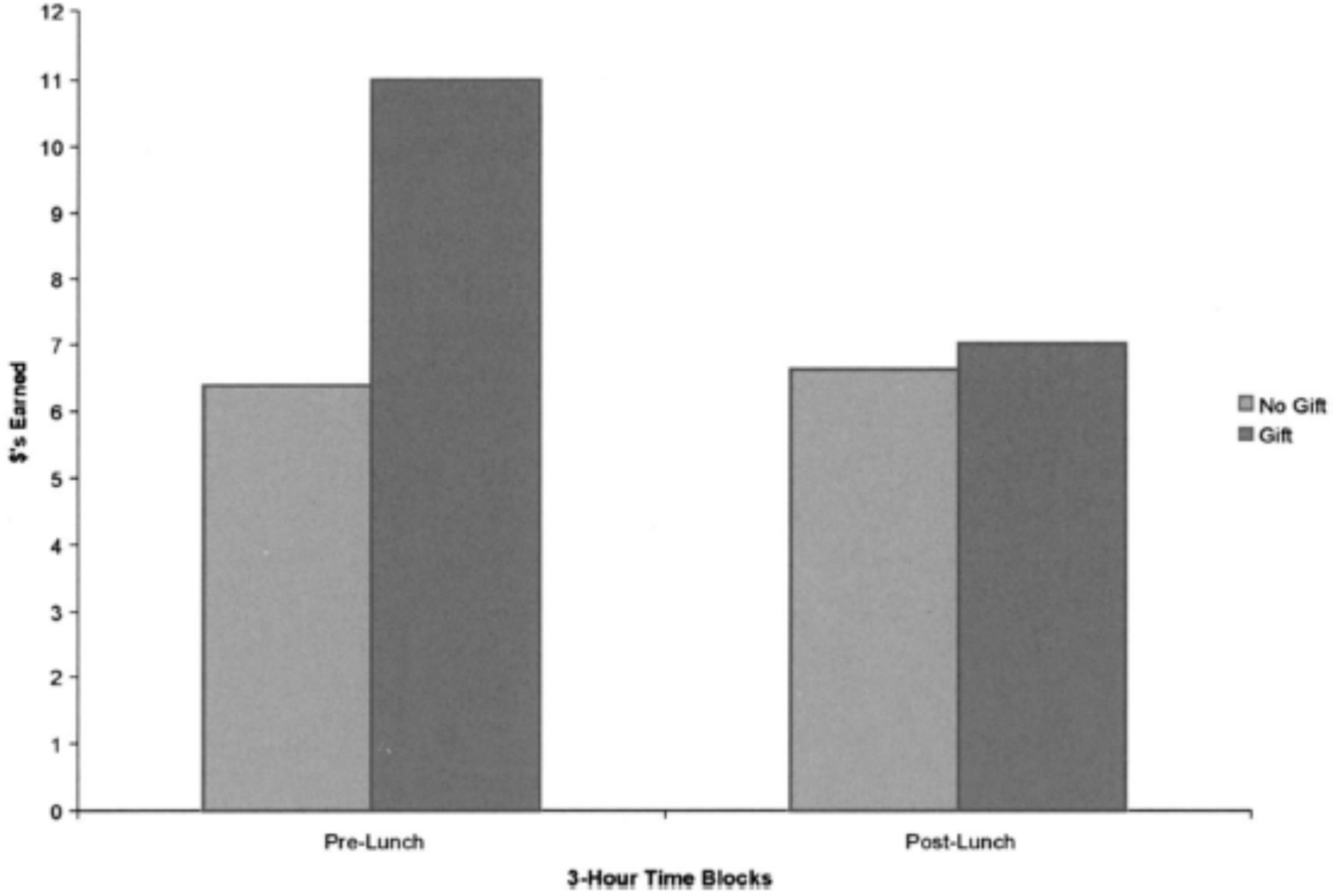


FIGURE 2.—Average earnings by 3-hour block.

Positive reciprocity on the labor market

Gneezy & List (2006)

- Interpretation
  - Reciprocity automatic, later people start to think more pragmatically or reference points are not static
  - Giving workers higher wages to motivate greater effort due to reciprocity not profit maximizing in this setting

- Why important? Is reciprocity on the labor market a dead concept?
  - Negative reciprocity?
  - Type of gift?

Negative reciprocity on the labor market

Kube, Marechal and Puppe (2006)

- Motivation
  - Positive vs. **negative** reciprocity (some literature suggest negative reciprocity is a stronger motive than positive reciprocity)
- Task
  - Cataloguing books for six hours, poster: *presumably* 15euro/hour
- Treatments
  - Control: 15euro/hour
  - Positive reciprocity treatment: 20euro/hour
  - Negative reciprocity treatment: 10 euro/hour

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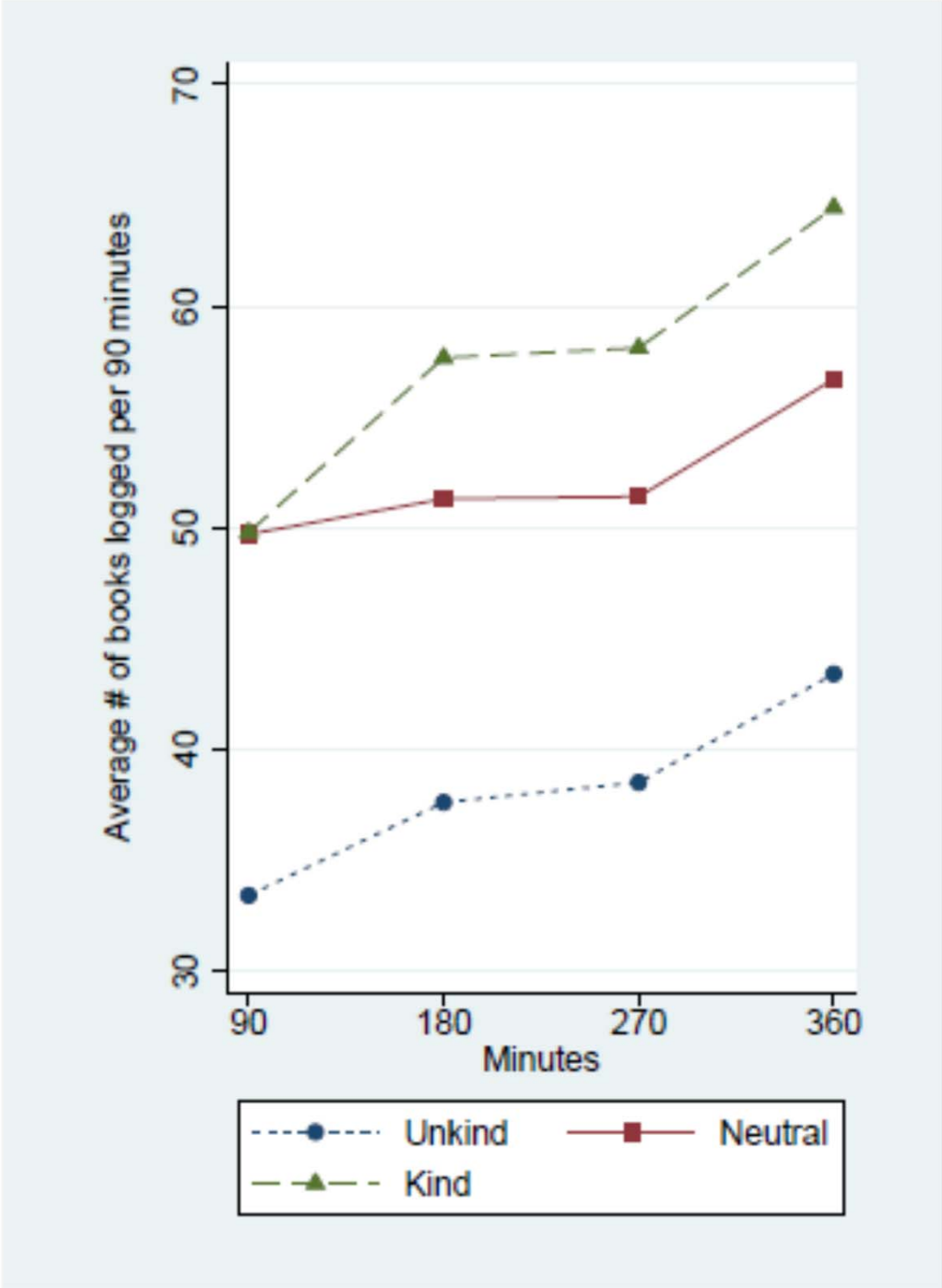


What effects do you think the authors found? How do people respond to a surprise cut in their wage?

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Negative reciprocity on the labor market

Lesson: Workers cuts cause persistent reduction of effort [also found in other studies]



The currency of reciprocity

Kube et al. (2011)

- Motivation
  - Previous experiments focused on the role of monetary gifts in triggering reciprocal responses of employees. Non-monetary gifts?
- Setting
  - Cataloguing books, no reemployment, using computer to monitor quality
  - Treatments
    - Control: The job was announced with hourly wage euro 12
    - Cash treatment: workers received a monetary gift of a 20% pay increase (euro7)
    - Bottle treatment: workers received a thermos bottle of equivalent monetary value
- Why a priori non-monetary gifts should affect productivity more than monetary ones?
  - …..
  - …..

The currency of reciprocity

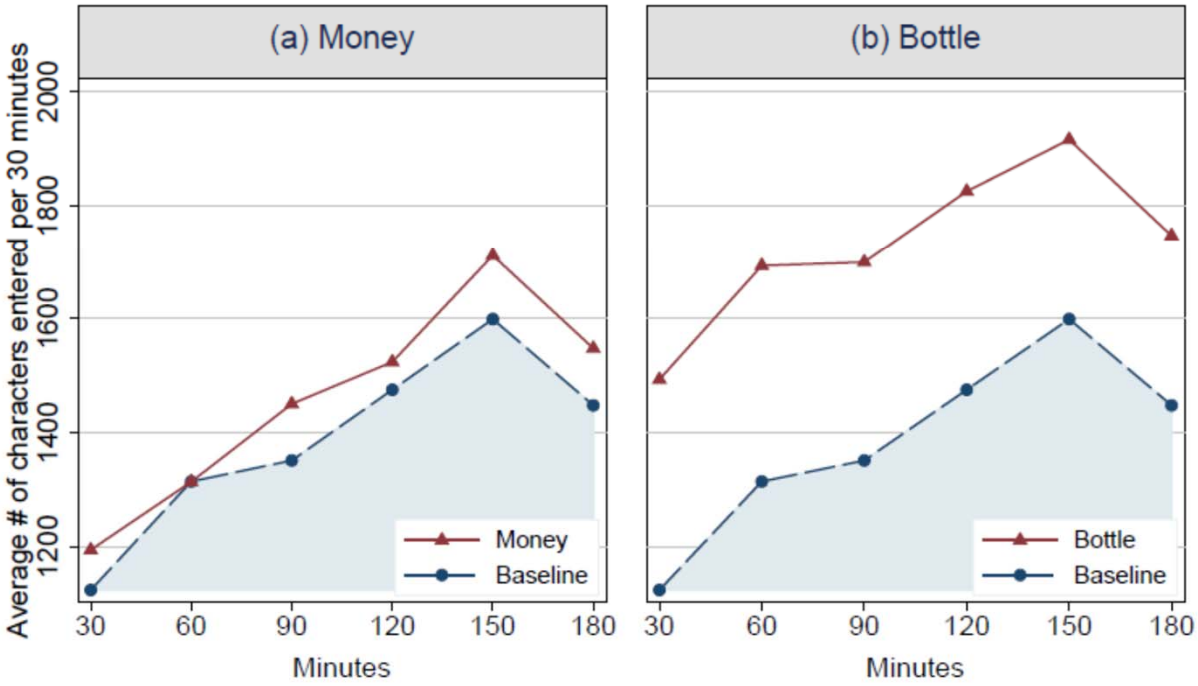
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    - Bottle treatment: workers received a thermos bottle of equivalent monetary value
- Why a priori non-monetary gifts should affect productivity more than monetary ones?
  - People value the “thought”
  - Preference for non-monetary gift
  - Beliefs about higher value (even in inaccurate)

The currency of reciprocity

- Cash does not increase effort, but bottle does

Figure 2: Money versus Bottle

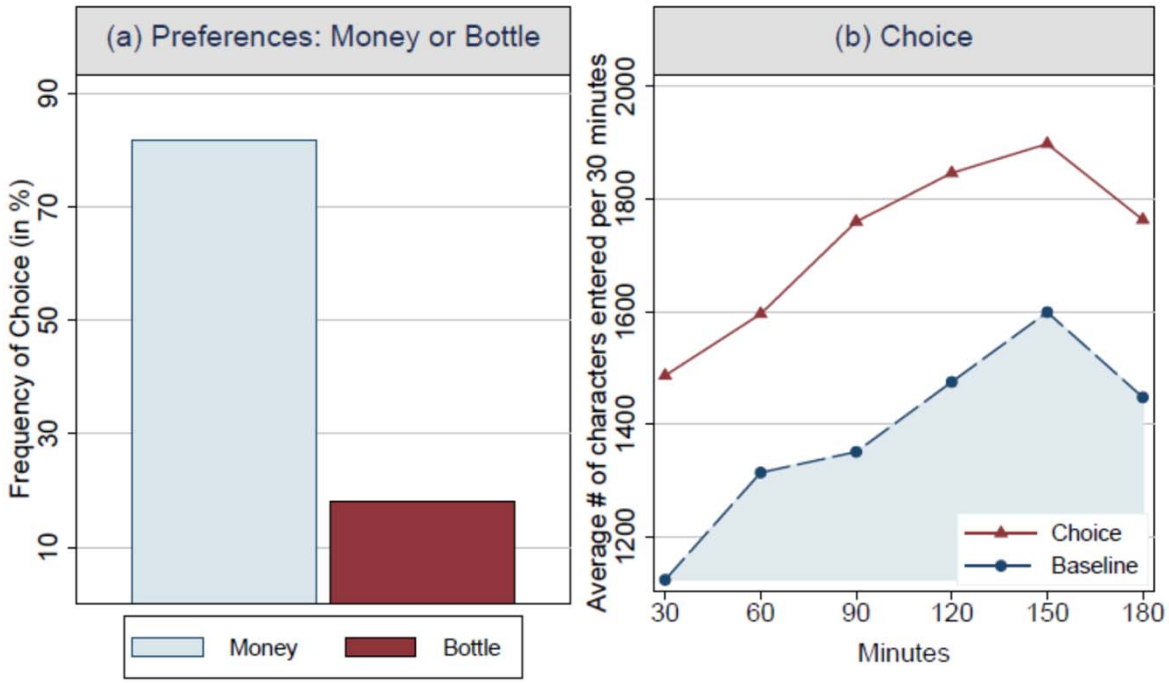


- Due to beliefs? No, additional treatment where participants were told the price of bottle

The currency of reciprocity

- Due to preference for non-monetary gifts?
  - New treatment: first choice b/w cash and bottle -> most participant took cash & but they still increase productivity

Figure 4: Choice



The currency of reciprocity

- Due to time and effort invested in the gift?
  - Origami treatment: employers gave the money in the form of origami – identical to the cash gift, but it was clear the employer invested effort
  - Effort increased by 30%, supporting the mechanism

Figure 1: Gifts In-Kind: Bottle and Origami



Notes: The first photo on the left depicts how the bottle was presented. In treatment PriceTag the €7 price tag was left visible at the bottom of the bottle. The other two photos contain the €7 origami in and outside of the envelope.

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- **Lesson:** It matters *how* bonuses are given to employees. Signaling genuine interest and care helps to trigger positive reciprocity.

Today

- Social motives on the labor market
  - Can employers motivate employees to exert more effort by giving them higher wages? Do wage cuts reduce effort? Does nature of the gift matters?
  - **Inequality in the workplace: Comparison with peers**

- Reciprocity in charitable giving: Can NGOs mobilize donations by distributing gifts?

Peers and creating **inequality** in the workplace

Breza, Kaur, Shamdasani (2017)

- **Does creation of inequality in the workplace reduce workers' effort?**

- Collaboration with Indian manufacturing firm (ropes, sticks, etc.)
  - Workers organized into production unites
  - Each unit sits together in a separate space
  - Production is individual activity (no joint production)

- Team randomly allocated into the following conditions
  - Condition 1: Same wages for all workers
  - Condition 2: Heterogeneous wages: each team member is paid a different wage
  - ➢ To identify the effect of relative pay on effort: comparing workers who receive the same absolute wage, but work with co-workers who receive different wages

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How do you think people respond to inequality in pay in the workplace? Do they increase effort if they get more compared to others? Do they reduce effort when they get less than others?

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Peers and impacts of inequality in the workplace

Breza, Kaur, Shamdasani (2017)

Finding 1

- Workers reduce output by 12% when their co-workers are paid more, 13% less likely to go to work
- No effects when co-workers are paid less
- ➢ Pay inequality demoralizes the weakest workers without boosting the strongest, and thus it is unambiguously bad for the firm.

Finding 2

- When workers are given a plausible reason for pay differences (due to baseline productivity levels), the negative effect on the lowest pay workers disappears
- ➢ Providing justification can attenuate demoralizing effects of inequality

Lesson: Unjustified wage inequality reduces effort

Today

- Social motives on the labor market
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- **Reciprocity in charitable giving: Can NGOs mobilize donations by distributing gifts?**

Reciprocity in charitable giving

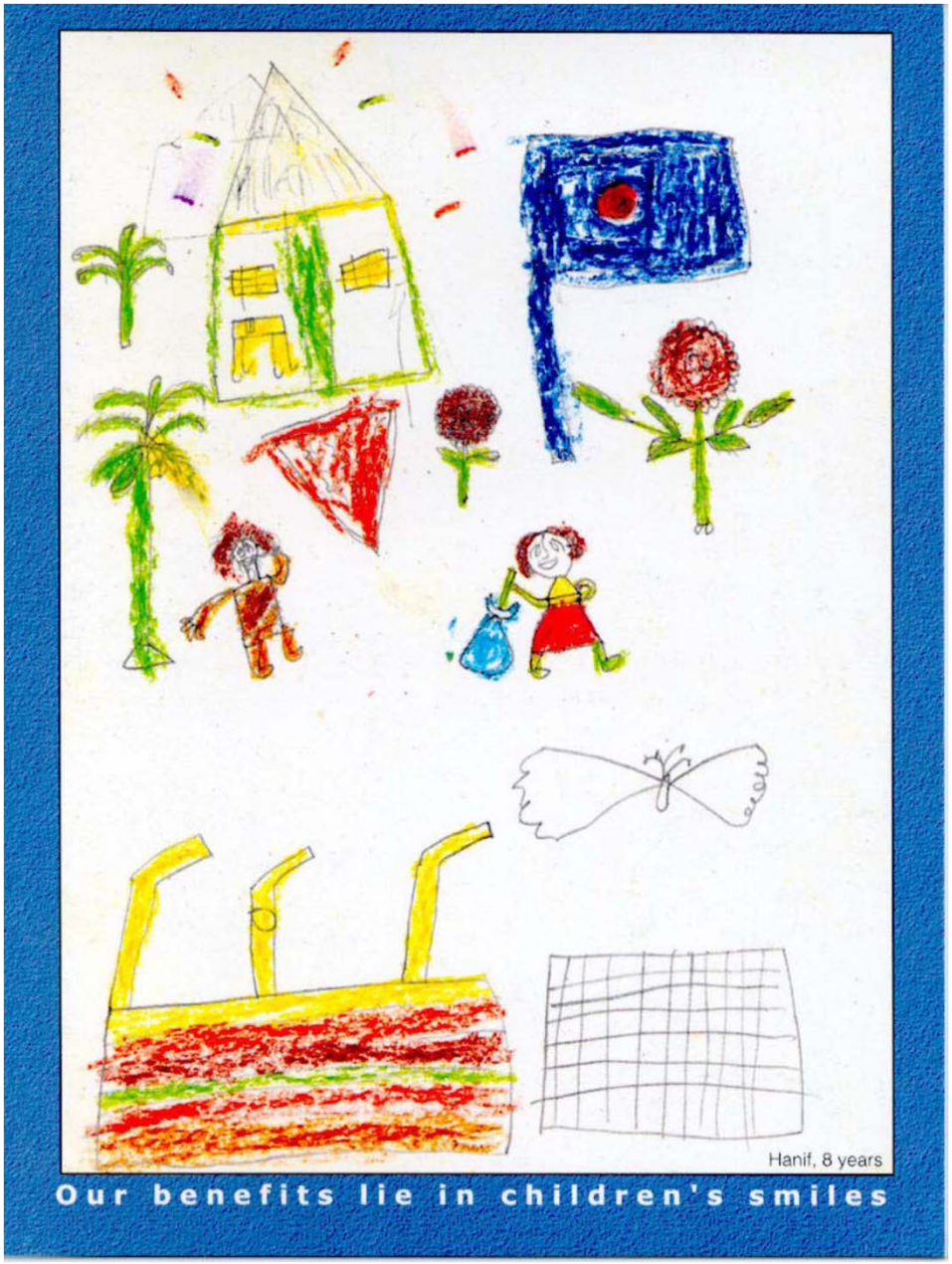
Falk (2007)

- Motivation
  - Can NGOs increase the amount of donations by taking seriously people's reciprocal preferences?

- Setting
  - Collaboration with an education NGO, 10 000 solicitation letters about a project, Switzerland->Bangladesh
  - Treatments
    - No gift condition (NGC)
    - Small gift condition (SGC): letter + postcard
    - Large gift condition (LGC): letter+ four postcards

- Predictions of theories
  - Selfishness: no donations
  - Altruism: unconditional donation (NGC=SGC=LGC)
  - Reciprocity: conditional donation (NGC<SGC<LGC)

Reciprocity in charitable giving



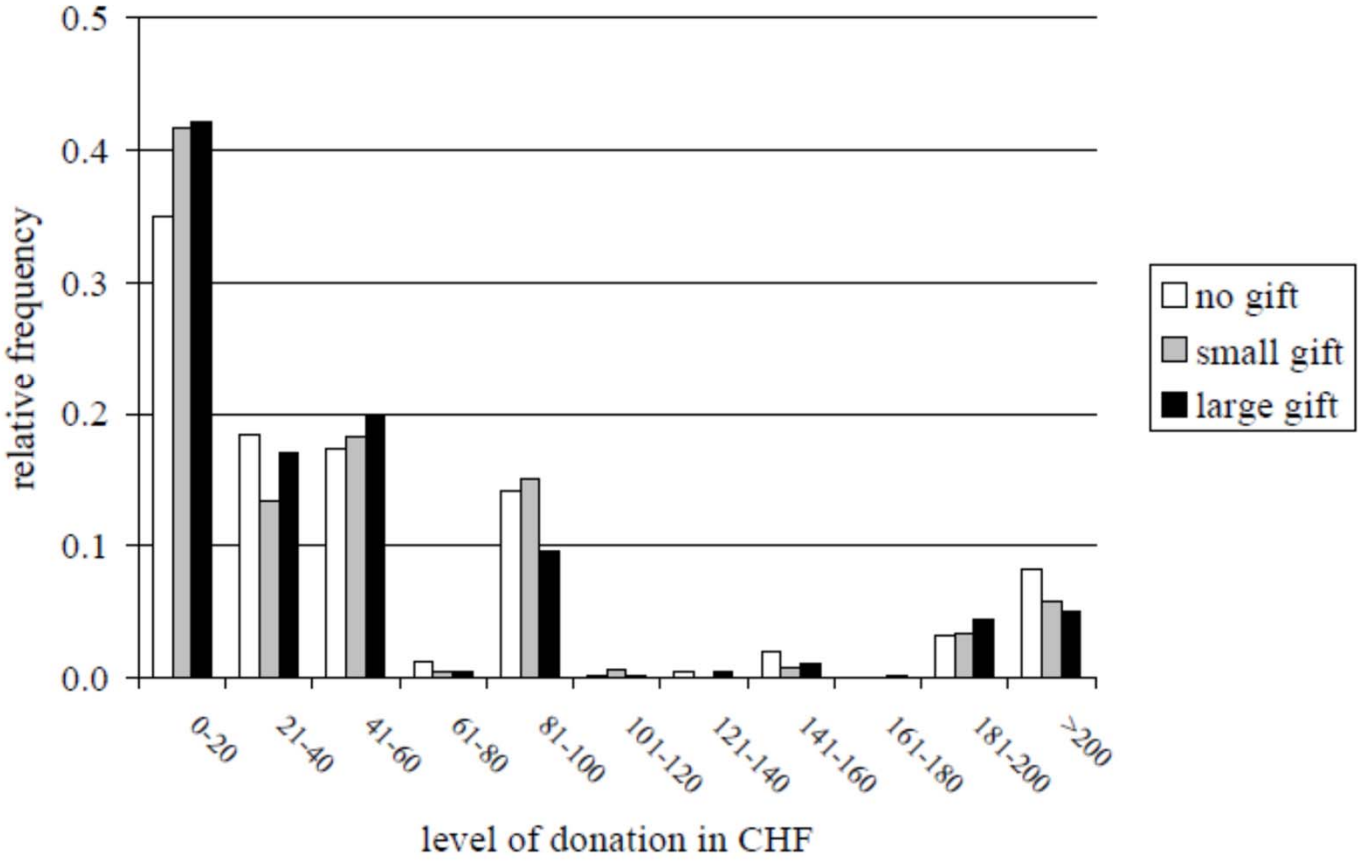
Reciprocity in charitable giving

TABLE 1: DONATION PATTERNS IN ALL TREATMENT CONDITIONS

	No gift	Small gift	Large gift
Number of solicitation letters	3,262	3,237	3,347
Number of donations	397	465	691
Relative frequency of donations	0.12	0.14	0.21

Reciprocity in charitable giving

FIGURE 1: HISTOGRAMS OF DONATIONS FOR EACH TREATMENT



Reciprocity in charitable giving

TABLE 3: ANALYSIS OF ABSOLUTE AMOUNTS OF DONATION AND POSSIBLE SUBSTITUTION EFFECTS

	No gift	Small gift	Large gift
Sum of donations *Christmas 2001 mailing* in CHF	24,673	27,106	40,877
Mean donation *Christmas 2001 mailing* in CHF	7.56	8.37	12.21
Sum of donations *February 2002 mailing* in CHF	14,023	13,206	13,165
Sum of *Christmas* and *February mailing* in CHF	38,696	40,312	54,042

Note: All donations smaller or equal CHF 500 (~ EURO 350)

Lesson: NGOs can boost donations by giving small gifts to donors

Today's lessons

- Social preferences matter
- Reciprocity and inequality aversion need to be taken into account when rewarding employees
  - Wage cuts reduce effort
  - Unjustified wage inequality reduces effort
  - Motivating employees by surprise bonuses might work, but it has to come hand-in-hand with a signal of care for the employee
- NGOs can boost donations by giving small gifts to donors (and thus triggering positively reciprocal responses)